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BeeQ: Evaluating Quantum Fidelity Kernels as a New Approach Methodology for Molecular Toxicity Prediction

MATCHED COMPARISON | 893 MOLECULES | 10 SIMULATED QUBITS

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AGENDA

01 **Context**

Why honey-bee toxicity matters, and the research question that drives the work.

02 **Methodology**

The dataset, how molecular families are separated, and how each molecule is represented.

03 **Implementation**

What a kernel is, how descriptors are encoded into quantum states, and the scope of the simulation.

04 **Results**

Performance across model families, transfer beyond Development, and the external CR8 panel.

05 **Conclusions**

What the study does and does not show, and where it goes next.

THE TEAM

It started at Quantathon 2026 CR





There are far more substances than feasible assays

01 · EXPOSURE

Apis mellifera is a key pollinator

The lethal dose is an established regulatory criterion.

02 · MEASUREMENT

Measuring it requires animal testing

Slow, costly, and it does not scale to the number of registered substances.

03 · PRIORITISATION

Not every molecule: the ones that matter

Models trained on already measured data allow the prioritisation of molecules.

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Not every molecule: the ones that matter

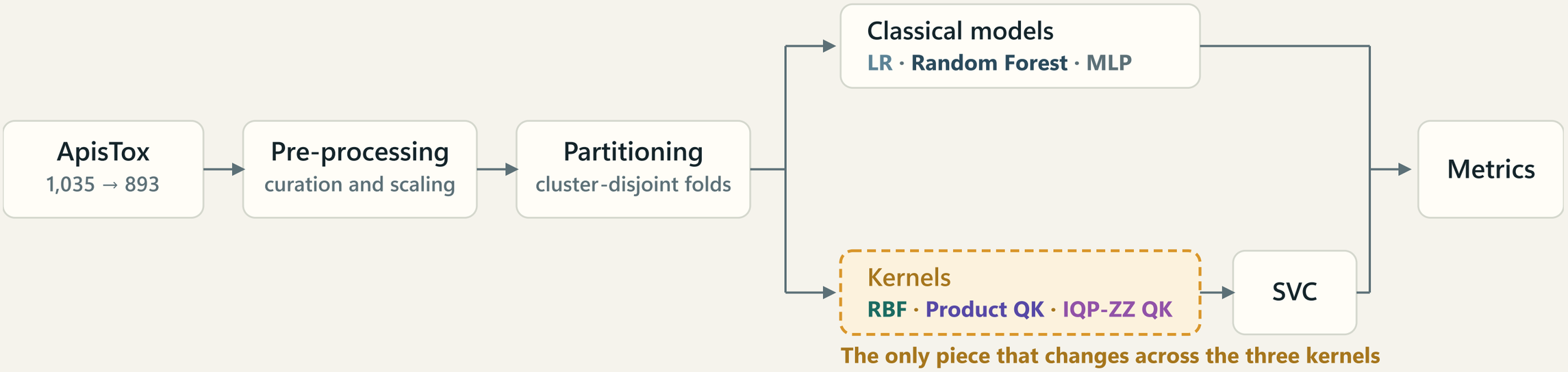
Models trained on already measured data allow the prioritisation of molecules.

BeeQ sits at the third step: an exploratory prioritisation tool, not a regulatory decision criterion.



RESEARCH QUESTION

Does a fixed quantum fidelity kernel induce a predictively and geometrically different similarity structure when the molecular representation, partitions, pre-processing and evaluation are kept unchanged?



--- Conceptual schematic

893 curated molecules with measured acute toxicity

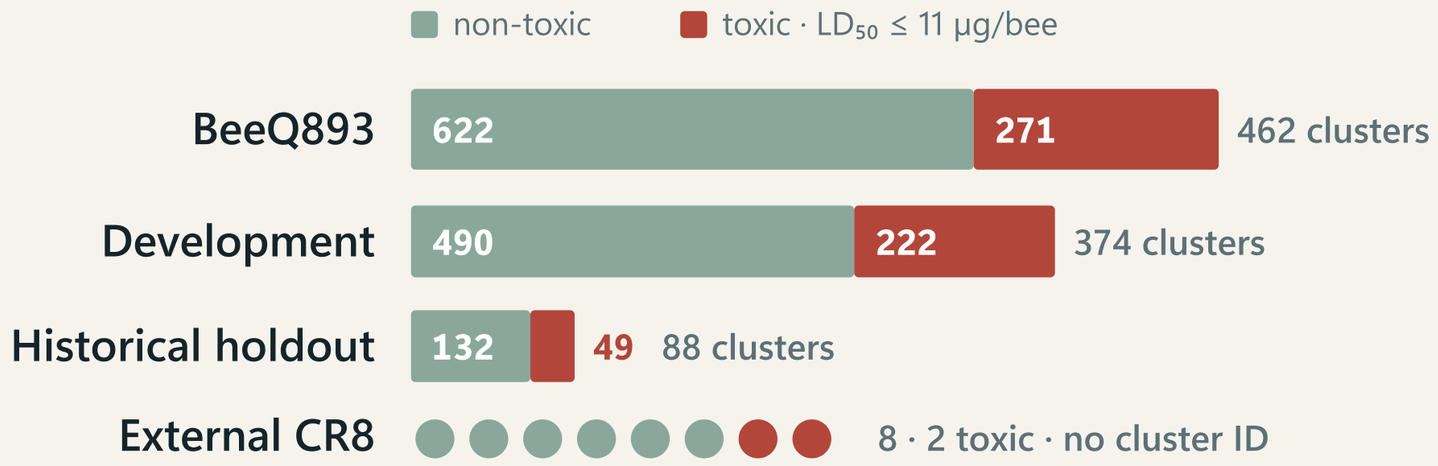
- **1,035 → 893 organic molecules**
  Metal-containing structures excluded.
- **Binary classification, not regression**
  The threshold was fixed before modelling.

THE ENDPOINT

$$y = 1 \text{ if } LD_{50} \leq 11 \text{ µg/bee}$$
$$y = 0 \text{ if } LD_{50} > 11 \text{ µg/bee}$$

— Threshold from ApisTox, following U.S. EPA · Health Canada PMRA · California DPR, *Guidance for Assessing Pesticide Risks to Bees* (2014)

893 curated molecules with measured acute toxicity



— SI Table S1 · frozen campaign

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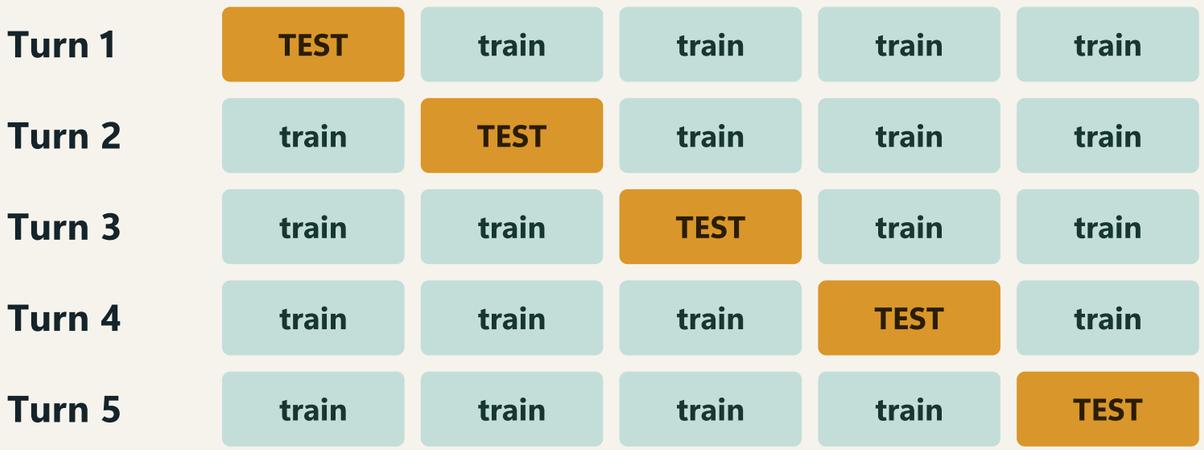
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Nothing that is evaluated takes part in its own selection

OUTER · Evaluate generalisation

Five turns: each molecular family is tested exactly once



INNER · Choose settings

Only from the outer training portion



4 inner folds · tuned on mean AUROC

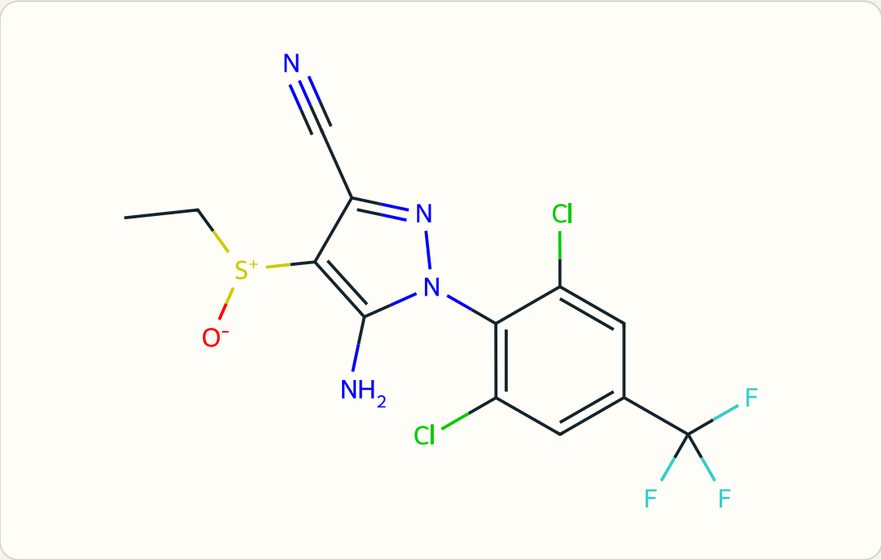
Holdout (181) stays untouched

Clusters stay together.

One outer fold tests each turn.

Morgan radius 2 · Tanimoto · Butina 0.60 → 462 clusters, 334 singletons, largest 40

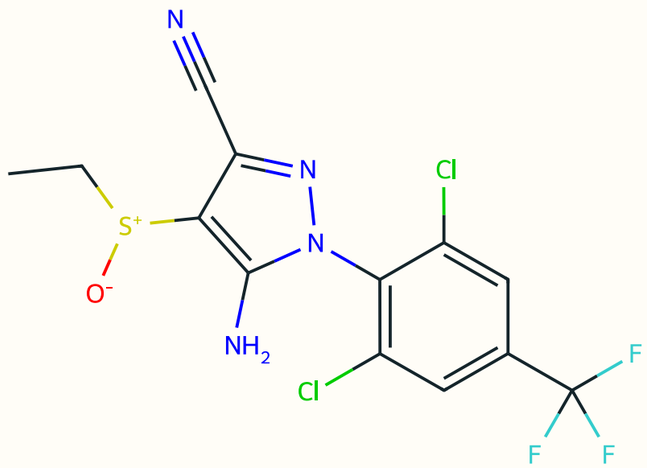
From a chemical structure to ten numbers



Ethiprole

— Structure drawn with RDKit from the frozen CR8 SMILES

From a chemical structure to ten numbers



Ethiprole

— Structure drawn with RDKit from the frozen CR8 SMILES

logP 1
3.78
z +0.44
θ +0.109

MW 2
397.21
z +0.77
θ +0.193

TPSA 3
90.69
z +0.37
θ +0.092

HBD 4
1
z +0.12
θ +0.031

RotB 5
3
z −0.52
θ −0.130

ArR 6
2
z +0.78
θ +0.195

nHal 7
5
z +1.94
θ +0.486

nOP 8
0
z −0.36
θ −0.090

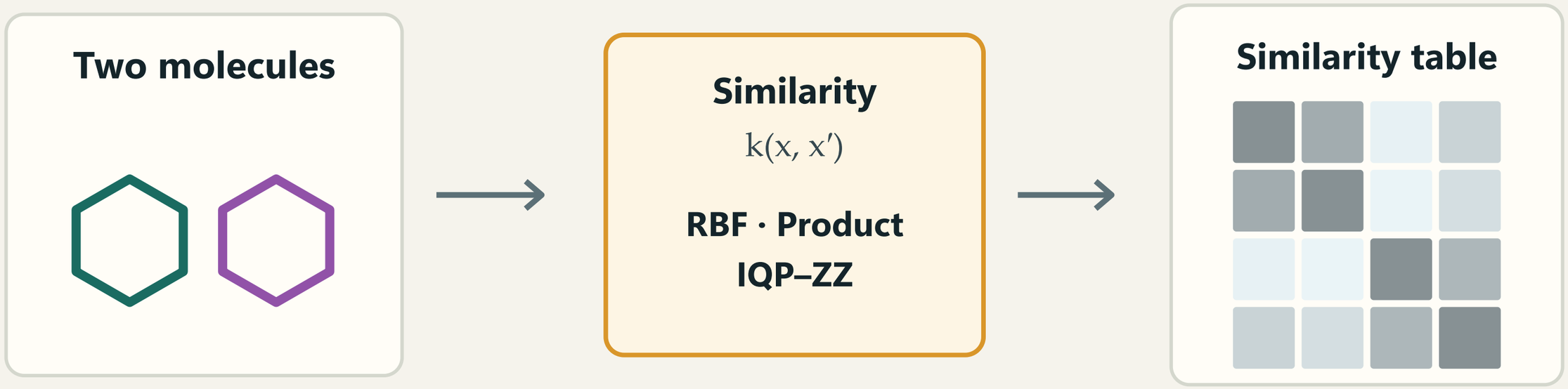
LiPHEX 9
−1.92
z −1.62
θ −0.406

fSASA 10
0.19
z +0.27
θ +0.067

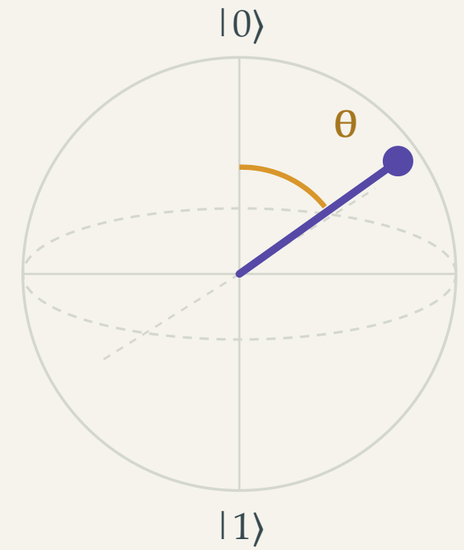
- **value** of the descriptor computed on the structure
- z **standardised:** (value − mean) / SD, with Development mean and SD
- θ **angle:** θ = s · z, what rotates the qubit, with s = 0.25

A kernel is a way of measuring likeness

Instead of asking the model to reason about each molecule on its own, we hand it a table with the likeness of **every pair** of molecules.

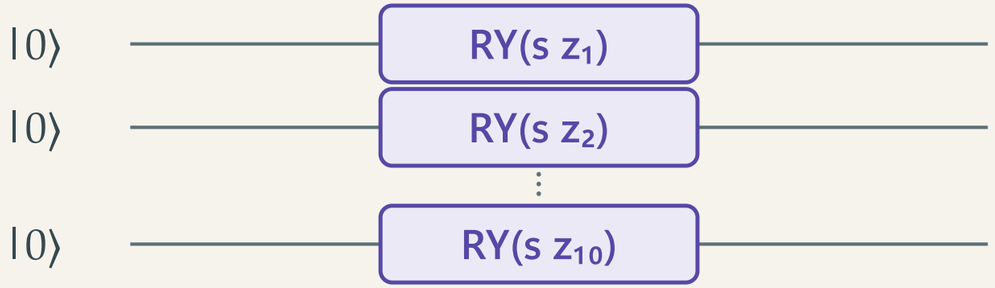


Each descriptor rotates a qubit; likeness is the fidelity



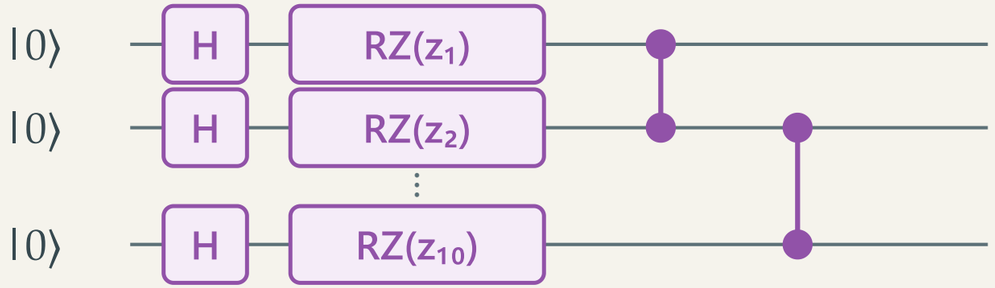
$\theta_i = s \cdot z_i \cdot$ one qubit per descriptor

PRODUCT · s = 0.25



no entanglement

IQP-ZZ · s = 0.125 · λ = 1



RZZ chain

nearest neighbours

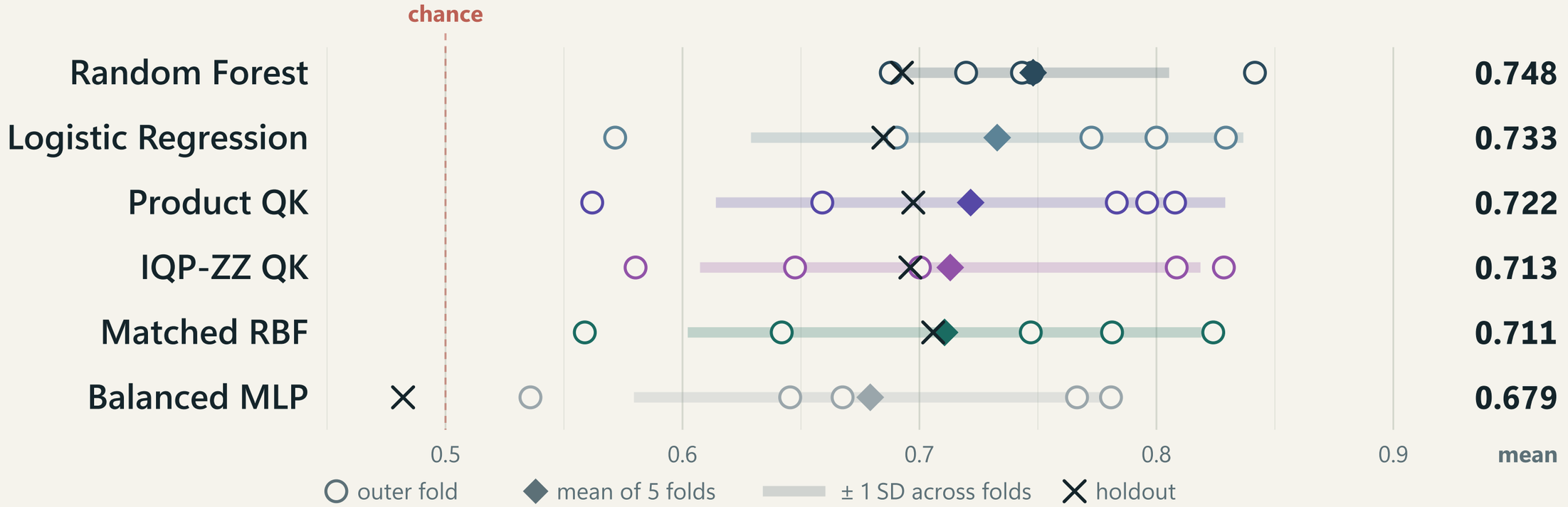
no final Hadamard

--- Conceptual schematics

- Exact statevectors in NumPy
- 1,024 amplitudes per molecule
- No shots, no noise, no quantum hardware

Dispersion across folds exceeds the differences between models

Dispersion across folds exceeds the differences between models



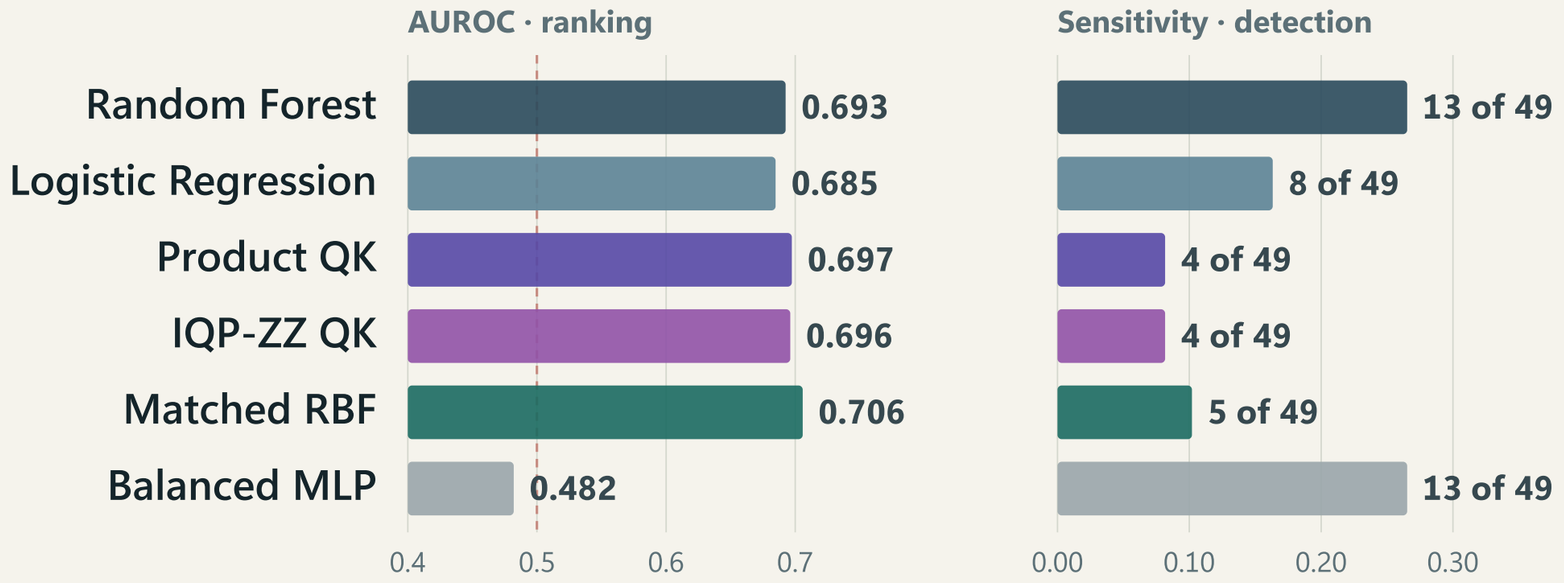
Random Forest reached the highest mean: 0.748

AUROC — ranking, not accuracy

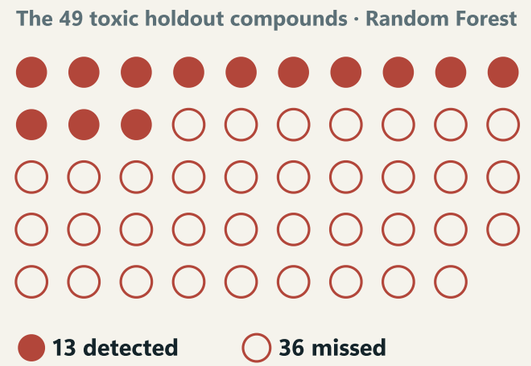
The probability that a random toxic molecule scores above a random non-toxic one.

— Paper Table I · SI Tables S17 and S18

Ranking well and detecting little are compatible

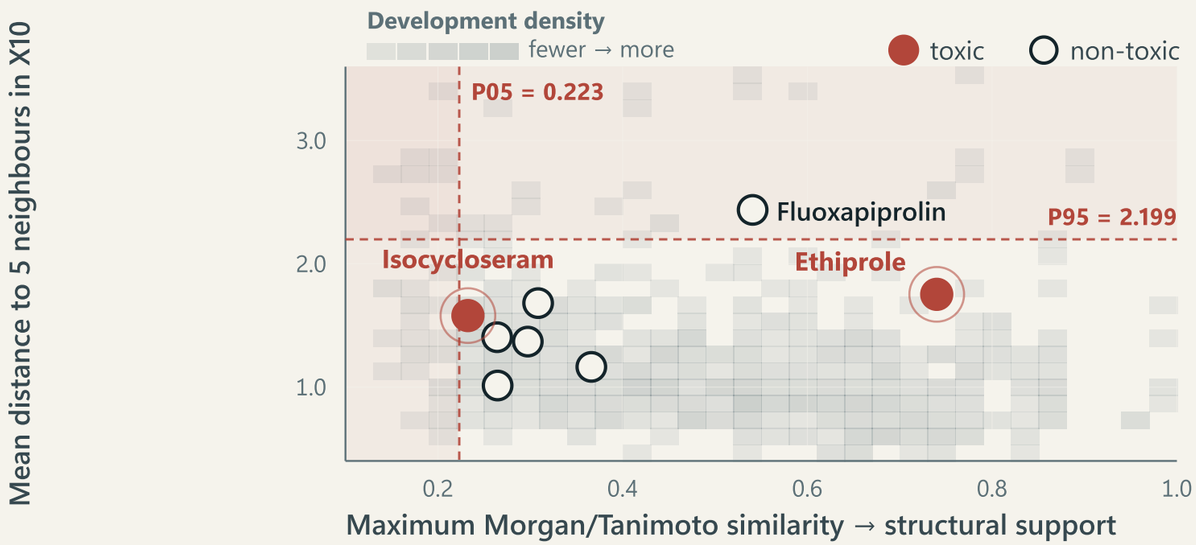


— SI Table S18



Five models hold AUROC near 0.70 and still let most of the toxic compounds through.

Being inside the domain does not guarantee a correct call



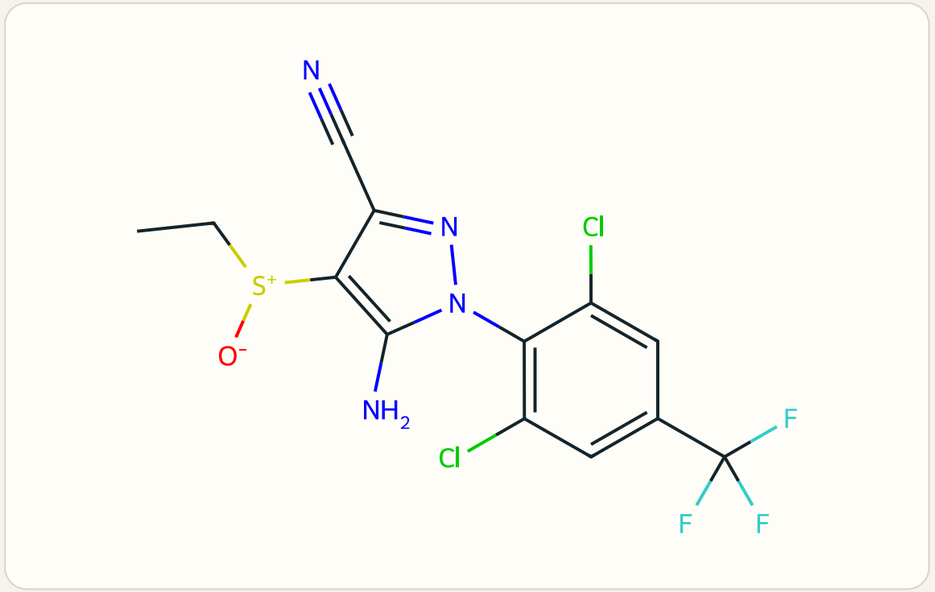
SI Table S26

	RF	LR	Prod	IQP	RBF	MLP
Isotianil	○	○	○	○	○	○
Fluoxapiprolin	○	○	○	○	○	✕
Ethiprole	✕	✕	✕	✕	✕	✕
Cyclobutrifluram	○	○	○	○	○	○
Isoprothiolane	○	○	○	○	○	○
Isocycloseram	✕	✕	✕	✕	✕	○
Spiropidion	○	○	○	○	○	○
Florylpicoxamid	○	○	○	○	○	✕

○ correct decision ✕ error frozen thresholds, no retuning

SI Table S27 · balanced MLP

Structural similarity to a toxic neighbour was not enough



Ethiprole

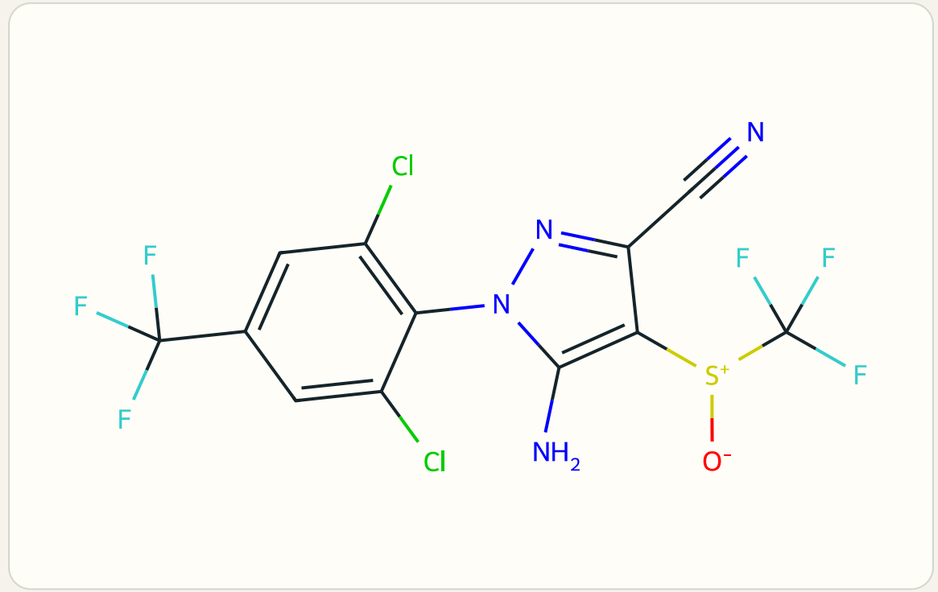
CR8 · external

● toxic

predicted non-toxic ×6

0.740

Morgan/Tanimoto similarity



Fipronil

Development · ID 387

● toxic

in training

Two findings, and a clear boundary

WHAT WE FOUND

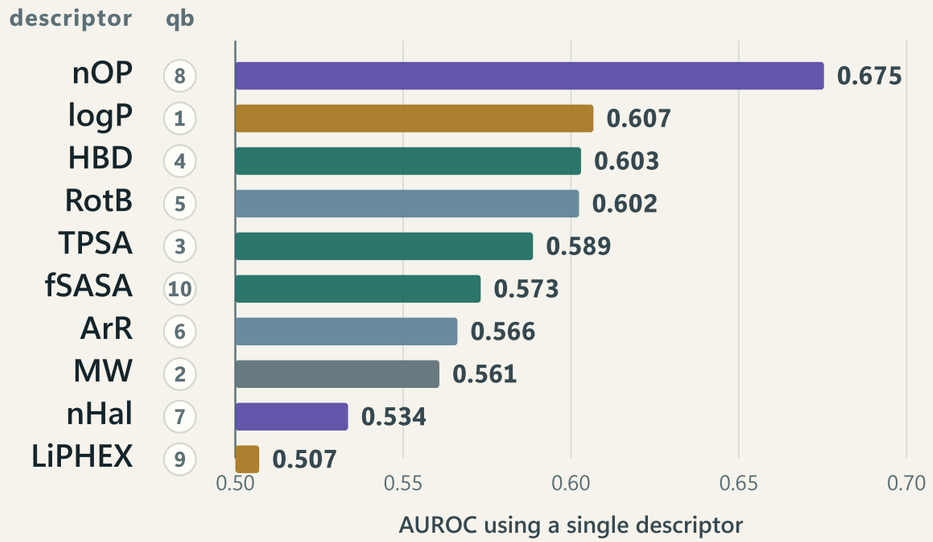
- **No consistent predictive improvement**
  Both fidelity kernels landed very close to the matched RBF.
- **No clearly distinct geometry**
  The Gram matrices turned out highly aligned with each other.
- **Detection remains the challenge**
  Beyond Development, sensitivity falls to 0.082–0.265.

WHAT THIS WORK DOES NOT SHOW

- It does not demonstrate quantum advantage
- It does not evaluate noise or real quantum hardware
- It does not establish clinical or regulatory utility

Separating kernel changes from representation changes

- **Optimise the encoding scale**
  Bandwidth turned out to be the decisive factor.
- **Train the embedding by kernel-target alignment**
  Instead of fixed maps, learn the map's parameters.
- **Rearrange and widen the qubits**
  The RZZ chain couples neighbours only: which descriptor sits next to which matters.



SI Table S5 · descriptive

Thank you.
Questions?

DATA AND CODE

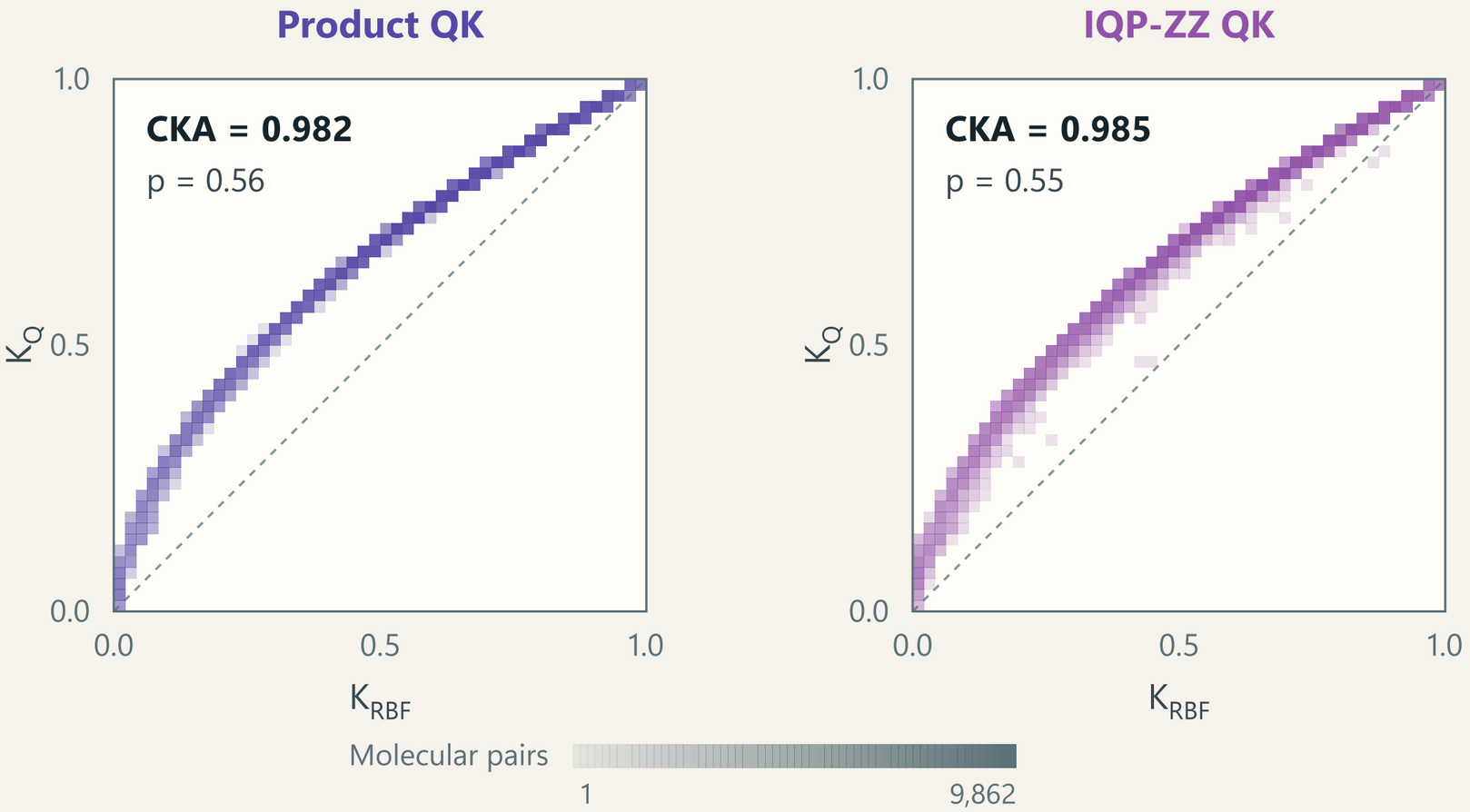
Supporting Information on Zenodo
doi.org/10.5281/zenodo.23329579

Dataset, descriptors and training workflow
github.com/andresquesadag/BeeQ

Scan for the repository

SUPPLEMENTARY KERNEL GEOMETRY

The two quantum kernels against the matched RBF



All 253,116 unique Development pairs

SUPPLEMENTARY X10 IN DETAIL

What each of the ten descriptors is

QUBIT	SYMBOL	COLUMN	DEFINITION	UNIT
1	**logP**	MolLogP	Lipophilicity: octanol/water partition estimated by the Wildman–Crippen method.	—
2	**MW**	MolWt	Average molecular weight.	g/mol
3	**TPSA**	TPSA_SP	Topological polar surface area, including sulfur and phosphorus contributions.	$Å^2$
4	**HBD**	NumHDonors	Hydrogen-bond donor count.	count
5	**RotB**	NumRotatableBonds	Rotatable-bond count: a proxy for molecular flexibility.	count
6	**ArR**	NumAromaticRings	Aromatic-ring count.	count
7	**nHal**	nHalogen	Number of F, Cl, Br and I atoms.	count
8	**nOP**	n_OP	Matches to an organophosphorus-like SMARTS motif. It is **not** the phosphorus count: several matches can involve the same atom.	count
9	**LiPHEX**	LiPHEX_prediction	n-hexadecane/water partition predicted by the in-house LiPHEX model: a more apolar lipophilicity regime than logP.	—
10	**fSASA**	sasa002_frac_polar_hetero_only	Fraction of the solvent-accessible surface assigned to polar heteroatoms (O, N, S, P): a 3-D measure of polar exposure.	fraction

The descriptor order fixes the qubit assignment